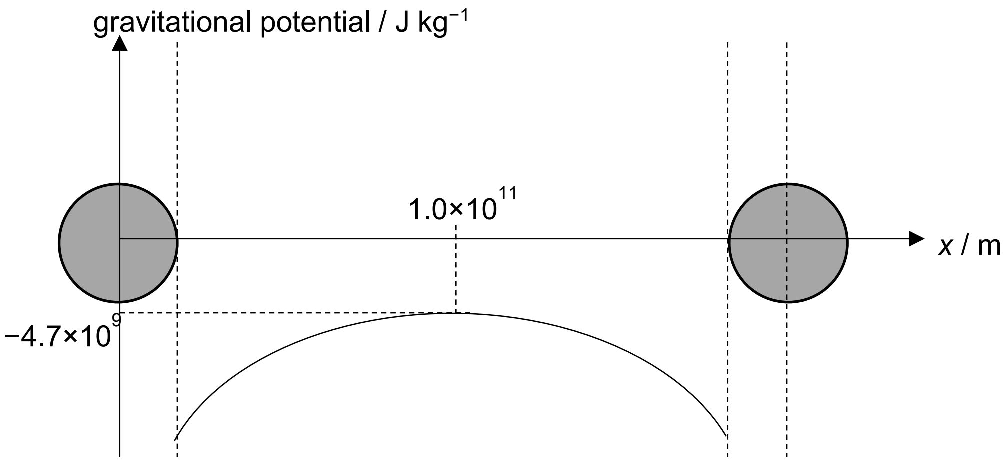

2ai	The gravitational potential at a point is the work done per unit mass (by an external force) in bringing a small test mass from infinity to that point.
2aii	gravitational potential at O = gravitational potential due to M at O + gravitational potential due to N at O $=-\frac{GM}{0.5d}+(-\frac{GM}{0.5d})$ where d is the distance between the centres of the stars $=-\frac{4GM}{d}$ $=-\frac{4(6.67\times10^{-11})(3.5\times10^{30})}{2.0\times10^{11}}$ $=-4.67\times10^{9}\ \text{J kg}^{-1}$

2aiii	For the asteroid to escape from the stars, it needs to reach infinity with min k.e. By principle of conservation of energy, **EITHER** total energy of asteroid at O = total energy of asteroid at infinity $m_{asteroid}\phi + \frac{1}{2}m_{asteroid}v_{\min}{}^2 = 0+0$ $\frac{1}{2}m_{asteroid}v_{\min}{}^2 = \frac{4GM}{d}m_{asteroid}$ $v_{\min} = \sqrt{\frac{8GM}{d}}$ $= \sqrt{\frac{8(6.67\times10^{-11})(3.5\times10^{30})}{2.0\times10^{11}}}$ $= 9.7\times10^{4}$ m s^{-1} **OR** loss in KE of the asteroid = gain in GPE of the asteroid $\frac{1}{2}m_{asteroid}v_{\min}{}^2 - 0 = 0 - m_{asteroid}\phi$ $\frac{1}{2}m_{asteroid}v_{\min}{}^2 = \frac{4GM}{d}m_{asteroid}$ $v_{\min} = \sqrt{\frac{8GM}{d}}$ $= \sqrt{\frac{8(6.67\times10^{-11})(3.5\times10^{30})}{2.0\times10^{11}}}$ $= 9.7\times10^{4}$ m s^{-1}
2aiv	Inverted U shape below horizontal axis Symmetrical 

2bi	The gravitational force is <u>just sufficient</u> to provide the centripetal force **OR** The acceleration/force and velocity of the stars are <u>constantly/always</u> perpendicular to each other
2bii	The gravitational force on the star provides the centripetal force $\frac{GMM}{d^2} = M(\frac{d}{2})\omega^2$ $\omega = \sqrt{\frac{2GM}{d^3}}$ $= \sqrt{\frac{2(6.67\times10^{-11})(3.5\times10^{30})}{(2.0\times10^{11})^3}}$ $= 2.416\times10^{-7}$ rad s^{-1} $T = \frac{2\pi}{\omega}$ $= \frac{2\pi}{2.416\times10^{-7}}$ $= 2.601\times10^{7}$ s $= 0.82$ years

3a	the *increase* in internal energy of a system is equal to the sum of the *heat supplied to* the